

KTH ROYAL INSTITUTE OF TECHNOLOGY
SCHOOL OF INDUSTRIAL ENGINEERING AND MANAGEMENT

A CDIO-Driven Method for Warehouse Layout and SOP Enhancement at ReMade in Sweden AB

Conducted within the ML2308 CDIO Course in
Sustainable Production Development at KTH

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1. Introduction

Warehouses play a key role in supply chains, where efficiency and accuracy directly affect overall system performance. As customer demand and product variety increases, warehouses should be able to handle higher throughput while maintaining reliability. Recent studies show that warehouse design and control is essential for identifying operational challenges and improvement opportunities in warehouse systems (Rouwenhorst et al., 2000; Zhen and Li, 2022).

ReMade in Sweden AB is a Swedish small-sized company that refurbishes and reuses IT equipment to extend device lifecycles, reduce electronic waste, and promote circular business. Key activities include structured handling, classification, and traceability of returned devices, thus warehouse operations are essential in enabling these circular-economy activities. The company operates under ISO 9001 (quality), ISO 14001 (environmental management), and ISO 27001 (information security), ensuring high-quality, secure, and environmentally responsible services.

As part of addressing these operational challenges, a project was conducted in collaboration with ReMade using Conceive-Design-Implement-Operate (CDIO) as a guiding framework. The project is carried out by four students from ML2308 HT25 CDIO course in Sustainable Production Development at KTH Royal Institute of Technology, with the objective to identify inefficiencies in warehouse workflows and explore possible improvements. This report follows the CDIO structure, organizing the work according to the Conceive, Design, Implement, and Operate phase.

2. Methodology

2.1. Overview of the CDIO Framework

The CDIO framework is a globally recognized framework for engineering education and project-based work, aimed at developing practical, real-world competence in system and process improvement (CDIO Standards 2.0, n.d.; Crawley et al., 2014).

- **Conceive** phase focuses on understanding stakeholder needs, defining project objectives, and establishing engineering requirements. During the project, initial analysis of ReMade warehouse workflow, identification of bottlenecks, and the setting of project goals were guided in this phase.
- **Design** phase focuses on translating identified problems and requirements into conceptual solutions. In this phase, several preliminary solutions were designed and evaluated, while selected solutions were further developed and refined to align with the objectives and constraints.
- **Implement** phase focuses on realizing the designed solutions. This phase involves applying proposed changes, testing functionalities, and validating their performance whether they satisfy the requirements and objectives.
- **Operate** phase focuses on supporting continued use of the implemented solution and improvement over time. This phase addresses monitoring, maintenance, lifecycle support and improvement mechanisms to ensure that operational gains are sustained.

By applying the CDIO framework, one can have a holistic structure for engineering projects, guiding problem identification, solution design, implementation, and operational evaluation.

2.2. Data Collection

In this project, various data collection methods were applied to ensure data quality and credibility. These methods enabled a comprehensive understanding of both process and human aspects of the operations:

- **Site visits and observations** – Direct observation of warehouse layout, workflow, and storage processes was conducted to identify bottlenecks and inefficiencies.
- **Operational data and internal documentation** – Warehouse inventory data, blueprints, and existing standard operating procedure were reviewed to understand current practices.
- **Surveys and feedback** – Surveys and feedback from operators and managers allowed team to capture qualitative insights on challenges, demands, and improvements effectively.
- **Interviews** – Interviews with managers and operators provided deeper understanding of problems and causes, allowing members to identify improvement opportunities more clearly.

2.3. Analytical Tools

To systematically analyze warehouse operations and evaluate improvement alternatives, established analytic tools were carefully selected and employed according to their purposes. The tools listed below provided a structured, evidence-based framework to assist decision making throughout the entire project.

Value Stream Mapping (VSM)

VSM was used to map material and information flows and identify non-value-adding activities (Abhishek & Pratap, 2020). Due to limited data availability and lack of digitalization in the warehouse, the analysis focused on observations and interviews rather than quantitative metrics.

Pugh Concept Selection Matrix (PCSM)

The PCSM systematically compared alternative solutions against a baseline, refining design options to best meet engineering requirements and practical constraints (Frey et al., 2009).

Multi-Criteria Decision Analysis (MCDA)

MCDA is applied to evaluate alternative layout solutions. Both qualitative and quantitative criteria could be considered by applying this tool. Solution is selected based on stakeholder priorities and operational requirements (Norheim, 2018).

Cost-Benefit Analysis (CBA)

CBA was used to evaluate the economic benefit of the proposed solutions. Expected costs and benefits were estimated based on observations and validation results, providing a financial evaluation of this project (Boardman et al., 2020).

2.4. Limitations

Several limitations were encountered during the project:

- **Data availability** – Historical operational data was limited to generate credible quantitative measures. As an alternative, qualitative methods are widely adopted during the project.

- Lack of digital maturity – Limited digital maturity in the warehouse operations restricted the implementation of advanced automation or digitalized solutions.
- Human resource constraints – The small team size combined with available skills limited the speed, scope, and complexity of data collection, analysis, and testing.
- Time constraints – The project duration (quarter year) limited both the extent of solution testing and monitoring of long-term impacts, and constrained the proposal of large-scale solutions.

2.5. Governance, Sustainability and Ethical Considerations

Sustainability and ethical considerations were addressed across all CDIO phases with different emphases. To ensure that these considerations are treated systematically aligning with the company's governance, a structured governance, sustainability and ethical consideration table was designed. Included considerations are listed below:

- **3 certificates that ReMade comply** – ISO 9001, 14001, and 27001
- **8 categories of ethical considerations** – 1. *Safety, health, and welfare of society*, 2. *Honesty, accuracy, and transparency*, 3. *Forward looking responsibility*, 4. *Openness to criticism*, 5. *Problem of many hands*, 6. *Maintaining technical competence*, 7. *Protection of sensitive information*, and 8. *Long-term societal impact*

The table is used as a checklist to verify that these requirements are considered consistently during the entire project. The governance-related components are derived by reviewing ISO standards that the company complies with, as well as relevant considerations from warehouse operations and general logistics practices. Ethical considerations are derived from the six ethical principles outlined in the IEEE Code of Ethics (IEEE, 2020), complemented by additional dimensions proposed in course material by Taebi (2021), resulting in eight ethical categories. Economic sustainability is considered in other sections using other tools, such as cost-benefit analysis, thus is ignored in this table. Full version and application of the table is presented in Section 7.

3. Conceive Phase

3.1. Project Mission

3.1.1. Objectives

The objective of this project is to support ReMade in identifying shortcomings in its warehouse operations and subsequently improving their efficiency, quality, and sustainability with solutions. The aim of the project is to not only provide short-term improvements, but also ensure that the implemented solution provides guidance to company operations in the long-term which contributes to the development of a scalable and efficient warehouse aligning with the company mission.

In line with this vision, the project mission is to develop sustainable, cost-effective, and scalable improvements that enhance process efficiency and quality while supporting the company mission by using the CDIO framework. To achieve these aims, the project scope is defined to ensure that the project

not only proposes practical improvements, but also develops a structured understanding of how warehouse performance can be improved in a sustainable way:

- Evaluating existing warehouse operations in a holistic perspective including layout, workflow, and digital maturity to identify inefficiencies and sustainability gaps, and
- Designing and validating a prototype solution in real operational conditions, and
- Proposing long-term lifecycle support practices to sustain solutions after the project ends.

3.1.2. Company Background

ReMade in Sweden AB is a sustainability-driven company that refurbishes and resells IT devices with the aim of extending product lifecycle and reducing electronic waste. The company operates according to a “Keep it Running” approach, which emphasizes device reuse, secure data erasure, and responsible recycling. To support these activities, ReMade follows established standards, ISO 9001, 14001, 27001, covering quality management, environmental responsibility, and information security.

The core business process includes acquiring used IT equipment from organizations and resellers, followed by selling refurbished devices through a structured grading and sales process. Operation functions between the two business processes and key activities include device collection, secure data wiping, testing, grading, and warehousing. While these processes ensure traceability and compliance of the products, operations rely heavily on manual labor and employee expertise, with limited digitalization and automation. As the company serves a diverse customer base and positions sustainability and reliability as key value propositions, inefficiencies in warehouses can lead to operational bottlenecks and variability in service quality. This highlights the importance of improving warehouse operations.

3.1.3. Business Strategy

Business strategy was analyzed by using information from company background and insights from site visits, interviews, and provided company documents. Balanced Scorecard (BSC) was chosen to analyze ReMade business strategy due to its holistic, multi-dimensional view of organizational performance, integrating four aspects; financial performance, customer value, internal processes, and learning and growth (Kaplan & Norton, 1992).

Given ReMade’s reliance on manual processes, employee experience, and sustainability-driven business model, this approach provides relevant insights, as summarized in Figure 1. From a financial and customer perspective, the analysis highlights ReMade’s strong sustainability-driven positioning which benefits from a reuse-oriented business model. The derived weaknesses provided hints to the team in identifying improvement opportunities. The internal process perspective, on the other hand, helped the team to pinpoint operational As-Is status. Finally, the learning and growth perspective gave the team insights about the way of working in the project, including how to communicate with the company and design of solutions and long-term sustainable practices to sustain solutions (Section 6.1.1 and 6.2.1).

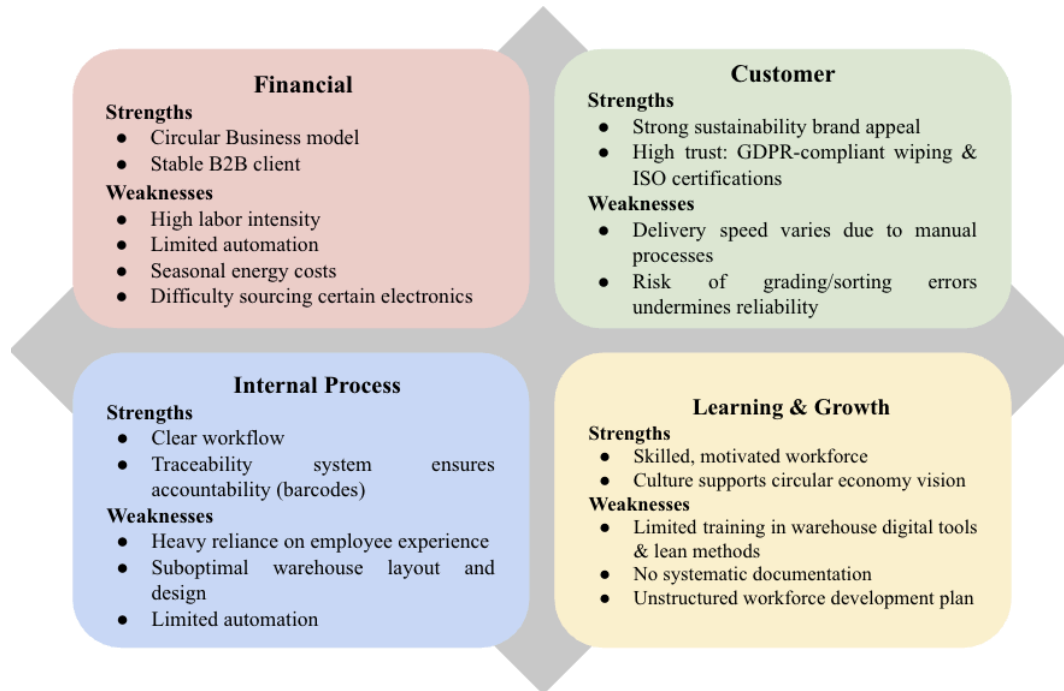


Figure 1. Balance Score Card

3.1.4. Stakeholder Analysis

A systematic stakeholder analysis was conducted to identify and understand the interests, needs, and impact levels of actors involved in or affected by warehouse operations. The analysis was based on on-site visits, semi-structured interviews, and direct observations. The scope of the analysis was defined according to the project's objectives and time constraints, in line with the principle that stakeholder identification should remain context-specific and purpose-driven (Freeman, 1984).

Stakeholders were categorized according to their influence on warehouse operations and their potential impact on proposed improvements. This classification follows established frameworks in stakeholder theories (Mitchell, Agle & Wood, 1997):

- **Low-influence** – Mainly external stakeholders including logistics partners and customers are included in this group due to their limited interaction and control of warehouse operations, resulting in minimum impact from proposed solutions in the warehouse (Siems et al., 2023).
- **Medium-influence** – The Production and the Buyer Team were classified as medium-influence stakeholders, since their activities directly affect warehouse performance. This reciprocal relationship positions them as relevant but not decisive stakeholders, but can influence operational intensity and workflow coordination (Agyabeng-Mensah et al., 2024).
- **High-influence** – High-influence stakeholders include individuals who have significant decision-making authority or are directly affected by operational processes. These stakeholders include: company manager, warehouse manager and warehouse staff. Warehouse staff are

especially critical stakeholders, as recent research highlights their essential experimental knowledge in actionable human-centered operational improvements (Tleuken et al., 2025).

	Warehouse Staff	Warehouse Manager	Business Manager	Production Team	Buyer Team
Role	Daily operations	Manage production and warehouse	Strategic decisions	Process items	Handle incoming orders
Influence	High	High	High	Medium	Medium
Interests	• Efficiency • Workload • Safety • Clear instructions • Communication	• On-time orders • Space utilization • Coordination with production	• Efficiency • Cost-effective • Scalability • Sustainability	• Smooth transfer • Clarity in warehouse processes	• Fast • Accurate shipping
Observations	• Manual tasks • Frequent errors • Layout issues • High physical strain • Communication	• Uneven pallet flow • Communication • Layout congestion during high workload • Lack of standardization	• Experience-based processes • Opportunities for warehouse process improvement and automation	• Random arrival timing blocks space • Lack of standards	• Order priorities issues • Blocked spaces

Table 1. Stakeholder Analysis

The information obtained from this process is summarized in the stakeholder analysis matrix (Table 1). This information contributed to identifying improvement opportunities and evaluating proposed operational changes. Using stakeholder analysis perspective ensures that improvement strategies address both strategic priorities and daily operational importance (Siems et al., 2023; Tleuken et al., 2025).

3.2. Conceptual Design

The conceptual design aims to build a high-level understanding of warehouse operations, identify inefficiencies, and structure the problem space before detailed solutions are developed. According to the CDIO framework (Crawley et al., 2014), this step emphasizes stakeholder needs, process mapping, and requirement identification to ensure that subsequent design and implementation activities are grounded in a thorough understanding of the current system.

3.2.1. Problem Identification and Target Problem Definition

Problems in the ReMade warehouse operations were identified through applying all methods mentioned in Section 2.2 except the surveys. The warehouse challenges could be analyzed from two complementary perspectives:

- **Thematic grouping** – This group is based on the nature of the problem (Process, Operational, Visibility/Layout, Error/Optimization; Appendix B.1). This grouping allowed the team to easily create potential solutions.
- **Impact grouping** – This group is based on operational consequences (Time, Time & Quality, Quality, Safety/Ergonomics; Appendix B.2). This grouping enabled to derive functional requirements in the engineering requirement (see Section 3.2.2).

The value stream mapping (VSM) was developed to identify non-value-adding activities in the process (Appendix B.3). Data were collected through direct warehouse observations, interviews with managers and operators, and collaborative workshops, focusing on laptops as a representative product. The mapping covered two processes: the in-shelving process and the order picking and packing process, revealing inefficiencies such as manual handling, waiting, and extensive searching. These issues primarily impact time and quality, with ergonomic concerns from repetitive tasks, providing a structured basis for workflow and layout improvements.

Since the project could not address all identified challenges, prioritization was necessary to maximize impact within limited time and resources. Due to the warehouse's low digital maturity and lack of quantitative data, interviews were conducted with the company manager and operators to gather their insights. The interviews revealed related but distinct concerns: operators emphasized reducing non-value-adding and repetitive tasks as well as physical strain, while the manager focused on operational inefficiencies and the lack of scientific analysis of current warehouse processes.

Based on these insights, the target problem of this project was defined as **reducing warehouse processing time and improving process quality**. There are two reasons for this decision:

- In-shelving and order picking & packing processes are the most time-consuming processes in the warehouse. As they function as primary bottlenecks in daily operations, reducing processing time will have a direct impact on overall throughput.
- Lost items, which are mainly caused by barcode misreading from non-warehouse operators, require extensive searching time and negatively impact outbound operations and customer reliability. Improving processing efficiency in these core tasks enable warehouse operators to take back control over the in-shelving process, thereby reducing lost items.

Although ReMade currently operates at a relatively small scale, existing inefficiencies are already evident. As future growth is expected to increase product variety and process complexity, unresolved inefficiencies may become embedded in the system. Therefore, the project emphasizes not only on short-term efficiency gains, but also introducing lean philosophy to support continuous improvement and long-term scalability.

3.2.2. Engineering Requirement

The engineering requirements were formulated through a structured transition from qualitative operational observations to formal system-level objectives for the project. In accordance with Dick, Hull & Jackson (2017), this process deliberately avoided premature solution selection and instead focused on understanding the existing operational reality and its systemic consequences. As illustrated in Figure 2, the requirements are organized in three hierarchies: business requirement (why), functional requirement (what), and technical requirement (how). The business requirement defines the overall objective of this project. The functional requirement represents the specific impact. Lastly, the technical requirement specifies how the determined impact is achieved.



Figure 2. Engineering requirement

The derivation of each requirement level follows a bottom-up analysis from observed problems in problem identification (Section 3.2.1), rather than from specified company requirements on this project. The business requirement of this project is derived from the project mission (Section 3.1.1), while four functional requirements are directly adopted from the impact categorization used during problem identification. The technical requirements are formulated by expressing observed problems as conditions that the system should solve. This approach differs from the established engineering requirement development procedure mentioned in Dick, Hull & Jackson (2017). However, this project did not start from well defined problems, rather defined from the observed stakeholder needs. Lack of visibility also limited specifying technical requirements into quantitative value. Nevertheless, the defined engineering requirement still played a critical role in selecting the most impactful solutions and evaluating the prototypes.

3.2.3. Project Planning

Project planning provides a structured basis for decision making under uncertainties. Siegel (2020) notes that schedule planning, cost analysis and risk analysis are the key to successful project planning. Schedule planning translates project scope into time-based activities, while cost analysis estimates resource requirements to determine project feasibility in cost dimension. Risk analysis identifies potential risks and uncertainties, allowing time to prepare and act on issues in advance before major failures occur. In this project, work breakdown structure (WBS) is first designed based on the CDIO framework (Appendix A.1), followed by risk analysis (Appendix A.2), and progress plan (Appendix A.3). Cost analysis is conducted after solution design, since the project was very vague in the beginning.

The WBS divides the project into structured work packages with clear objectives and interconnections. Due to the vague problem definition, risk analysis was critically important. A risk breakdown structure (RBS) assessed likelihood and severity to generate risk scores. The progress plan integrates WBS and RBS to allocate time to each work package based on workload and risk. Progress and delays were tracked and shared among team members, with delays visualized using red indicators. The final version is presented in the report.

4. Design Phase

4.1. Preliminary Design

After careful analysis of the existing warehouse operations, four potential solutions were developed, followed by a solution selection step to identify the solution that best aligns with the project objectives.

4.1.1. Solution Candidates

4.1.1.1. *Layout Improvement*

Several problems that were mentioned in Section 3.2.1 were associated with inefficient warehouse layout design. Drawing a full warehouse layout (Appendix C.1) allowed the team to further pinpoint the hotspot areas that have highest improvement potentials. These issues include inefficient material flows, high physical strains, and high cycle time. Therefore, layout improvement is suggested to improve operational efficiency and ergonomics. This solution focuses on three main principles: optimizing shelf location, improving use of lower shelves and enhancing shelf numbering. Together, these aim to reduce unnecessary travel, improve picking accuracy and establish a more structured and standardized warehouse layout. In the short-term, the proposed solution is expected to minimize searching, traveling, and picking and packing time, and lessens physical strain, leading to higher throughput and operator satisfaction. In the long-term, this solution establishes a consistent and efficient workflow, and creates a foundation for future automation opportunities, while improved ergonomics foster a more sustainable work environment.

4.1.1.2. *Digital Dashboard*

The digital dashboard addresses challenges in slot visibility and inefficient item retrieval by integrating ERP data through an API, processing it via an ETL pipeline, and visualizing lane-level occupancy and item locations in Power BI. In the short term, it reduces time spent searching for storage and locating items, improves detection of misplaced items, and lowers physical strain for operators. In the long term, it prevents misplacement, enhances inventory accuracy and tracking, increases layout awareness, supports data-driven decision-making, and contributes to greater operator satisfaction (Appendix B.4).

4.1.1.3. *Standard Operating Procedure*

Standard Operating Procedure (SOP) is a set of detailed and standardized written instructions designed to achieve uniform task execution and consistent process quality of a specific task (Manghani, 2011). SOP in the project may not directly impact the processing time, but it mainly targets the time of assisting non-experts and reducing lost items. Since searching for lost items requires significant time and often disrupts downstream processes, such as order fulfillment and customer satisfaction, reducing lost items can lead to a substantial reduction in item searching time. In the longer term, SOP is also expected to improve process visibility, providing a capability for continuous improvement as operational complexity increases.

4.1.1.4. *Schedule Optimization*

Digital task scheduler is designed as a schedule optimization mechanism to address inefficient and experience-based task coordination in warehouse operations. Rather than functioning as a simple digital

task list, the system continuously prioritizes and assigns tasks based on real-time conditions such as task urgency, proximity, and operator workload. The main challenge targeted by this solution is inefficient task organization caused by frequent interruptions, uneven workload distribution, and lack of task visibility. In the short term, digital task scheduling is expected to indirectly reduce overall processing time by organizing tasks more efficiently, minimizing unnecessary task switching, and balancing workload within the team (Zhen & Li, 2022). In the long-term, the system is also expected to improve process visibility, operator satisfaction, and communication between warehouse and production teams, while enabling continuous optimization through using historical data. The schematic system flow is presented in Appendix B.5 including physical and system components.

4.1.2. Solution Evaluation

In this phase, the four proposed solutions are evaluated from multiple perspectives in order to identify the most effective solution combination for this project. The evaluation considered engineering requirements and cost analysis that can represent the impact of solutions and technical/resource feasibility respectively. For simplicity, the solutions are referred to as Solution I, II, III, IV, following the order presented in Section 4.1.1.

4.1.2.1. Satisfaction of Engineering Requirement

The solution should also be able to address all engineering requirements generated from the stakeholders. From the expected impact of the solutions mentioned in Section 4.1.1, Table 2 is created by connecting expected impact to the engineering requirements. This is later applied in Pugh Concept Selection Matrix to evaluate solutions' engineering requirement fulfillments and proceed with the best solution combination.

Time (Target)	Quality
- Reducing item searching time (I, II, III, IV) - Optimizing layout to minimize traveling time (I) - Reduce training time (III)	- Reducing shelving errors (II, III) - Standardize the process (III, IV) - Improving warehouse visibility to facilitate inter-team communications (I, II, IV)
Safety	Cost
- Designated space for packaging activities (I) - Minimizing items on the bottom shelf (I) - Minimizing manual input	- Increase item visibility to eliminate lost products (I, II) - Minimize non-value adding time (I, II, III, IV)

Table 2. Expected engineering requirement fulfilled by the proposed solutions

4.1.2.2. Cost Analysis

The four proposed solutions differ in complexity, implementation effort, and cost, as summarized in Table 3. Labor costs were calculated based on four team members working 4 hours per day at 200 kr per person per hour, following the estimates in the Pugh Concept Selection Matrix. Solution I and III are low-cost, quick-to-implement options, primarily requiring labor and minor materials or documentation. Solution II represents a medium-cost option, involving software, hardware, and ERP integration to

provide long-term operational visibility. Solution IV is the most complex and costly, requiring significant development and technical resources while offering advanced automation and optimization.

	Solution I	Solution II	Solution III	Solution IV
Labor	5 days of planning + 2-3 days of execution	10 days of planning + 15 days of execution	5 days of planning + 1 day of execution	30 days of planning and execution
System component	1,000 – 2,000 kr (labels, barcode stickers, etc.)	3300 kr for Power BI monthly subscription	~500 kr (documents)	~20,400 kr (scanners, tablets, storage, etc.)
Total cost	27,100 kr	80,330 kr	19,700 kr	100,400 kr

Table 3. Cost Analysis of Proposed Solutions

4.1.2.3. Pugh Concept Selection Matrix

To derive the best solution/solution combination that can give the highest impact with minimum cost, PCSM is applied. This matrix allows one to compare multi-criteria analysis between different candidates. Three aspects are considered: impact, cost, and feasibility. The impact is translated from the engineering requirement fulfillment. Cost analysis covers the cost of all resources including time and materials. Labor time for design and implement phases are also separately considered to measure the feasibility.

		Solution variations									
		<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>I + II</i>	<i>I + III</i>	<i>I + IV</i>	<i>II + III</i>	<i>II + IV</i>	<i>III + IV</i>
E.R.F.	Time	2	1	2	1	2	3	2	2	1	2
	Quality	1	3	4	2	3	4	2	4	4	4
	Safety	2	0	0	0	2	2	2	0	0	0
	Cost	2	2	1	1	2	2	2	2	2	1
Cost (SEK)		27.1K	80.33K	19.7K	100.4K	107.43K	46.8K	127.5K	100.03K	180.73K	120.1K
Labor (hrs)	Design	80	160	80	320	240	160	400	240	480	400
	Imp.	48	240	16	160	288	64	208	256	400	176

Table 4. Pugh concept selection matrix of the solution and combinations

Table 4 presents the result of the matrix. Red cells indicate the infeasibility. From the engineering requirement fulfillment (E.R.F.), solution II, III, IV, II+III, II+IV, and III+IV are infeasible since they do not satisfy safety requirements. Also, IV, I+II, I+IV, II+III, II+IV, and III+IV requires more time than dedicated time for each phase (see Appendix A.3). Therefore, solution I and I+III remain to be the final candidate, and I+III is chosen to provide the biggest impact with available resource constraints.

4.2. Detailed Design

4.2.1. Design of Layout Improvement

4.2.1.1. Layout Candidates

Alternative layout designs (A1–A6) were developed based on insights from warehouse operators and managers, reflecting solution principles listed in Section 4.1.1.1. Table 5 summarizes the key advantages and limitations of each alternative layout. The proposed layout designs (A1–A6) were evaluated against core requirements of cost, safety, quality, and time (Section 3.2.2). While A1–A3 focus on workflow efficiency and brand organization, A4 improves ergonomics and retrieval via the Golden zone principle,

and A5–A6 further optimize storage and handling efficiency. Appendix C.2. presents the detailed layout design of each alternative.

	Key Changes	Key Limitations
A1	Clear separation of inbound/outbound areas; U-shaped workflow; centralized packing materials.	May require temporary disruption of operations.
A2	Reduced noise; flexible shelf allocation by brand; improved operator environment.	Brand zoning is slightly less intuitive than A1.
A3	One aisle per brand; consolidated accessory storage; chargers arranged by brand/voltage.	Minor adjustments needed for packing-material placement.
A4	Golden Zone optimization (De Koster et al., 2007).; ergonomic improvements; S-shape routing reduces walking distance; modular/flexible storage.	Requires staff training; rebuilding frames and shelves increases labor demand.
A5	Increased shelf dimensions enhance storage capacity and accessibility.	Physical intensive labor and a lot of time to adjust shelf locations.
A6	Optimized box gaps and slot density; maximized storage efficiency; maintains ergonomic access.	Excessive planning time and extensive shelving modifications.

Table 5. Summary of Proposed Alternative Layouts (A1–A6)

4.2.1.2. Multi-Criteria Decision Analysis

To identify the most suitable layout design, MCDA was conducted for all six proposed alternatives. This structured approach enabled a comparison based on performance criteria and feasibility, ensuring alignment with operational objectives and practical constraints (Norheim, 2018). In this project, all layout designs are evaluated using four functional requirements in the engineering requirement (see Appendix C.3). Weights for each requirement, optimality, and feasibility scores are assessed qualitatively, ranging (0 - 1), (1 - 5), and (0 - 100) respectively. The result shown in Figure 3 indicates that A3 demonstrates the best overall alignment with both engineering requirements and operational feasibility. Accordingly, this layout design was selected for subsequent phases.

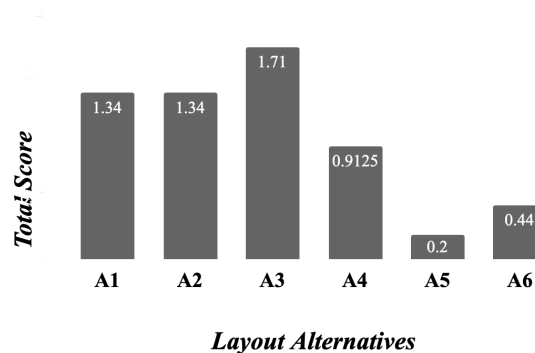


Figure 3. MCDA Scores of Layout Alternatives (A1–A6)

4.2.2. Design of Standard Operating Procedure

The detailed design of the SOP followed the diagram in Figure 4, which follows a typical iterative process. First, SOP document contents are determined by referring to scientific papers about SOP development guidelines. General structure, including clarity of purpose, standardized structure, defined responsibilities, and document control, were adopted from the studies (Hollmann et al., 2020; Barbé et al., 2016). Although these guidelines were originally developed for laboratory and quality management contexts, their logic still remains general and transferable to other domains, ensuring consistency and readability.

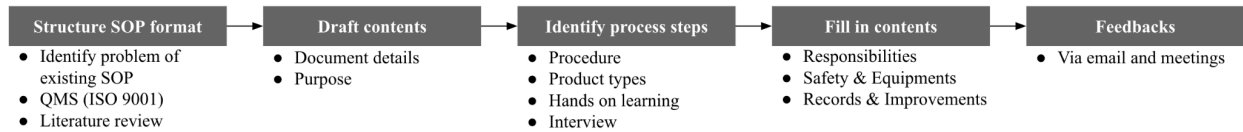


Figure 4. Detailed design procedure of SOP

Each section's purpose and contents were defined through discussions with the company, which revealed that existing SOPs were underused because the instructions were unclear. Level of detail and the inclusion of pictures were carefully considered for clarity and data privacy. Animated images generated by AI tools were used to comply with regulations. Procedures vary by product type, but laptops were selected as the primary focus due to time constraints. Detailed instructions, cautions, potential errors, and mitigation strategies were developed under guidance from warehouse operators. Two rounds of company feedback were conducted, first on the content draft and then on the completed SOP, with mostly positive responses.

5. Implement Phase

According to the CDIO framework (Crawley et al., 2014), the implement phase emphasizes the iterative process in realization of the proposed design to ensure that the solution is feasible, functional, and aligned with real operational conditions. The team mainly applied the Plan-Do-Check-Act (PDCA) cycle approach during this phase, by actively involving warehouse operators and managers in prototyping, receiving feedback and revising accordingly.

5.1. Implementation of Layout Improvement

5.1.1. Element Creation

For realization of the selected layout design, six coordinated domains are translated, justified, and executed during the first phase (element creation) of the Implement phase. This phase mainly focuses on creating physical components needed in the new layout design.

- **Layout components:** Installed aisles, shelves, and shelf levels; introduced a box labeling system to improve traceability and reduce pick errors.
- **Physical preparation:** Optimized box space, printed and applied brand/voltage labels, added floor markings and zone signs, and relocated physical units (e.g., chargers, packing tables).

- **Digital tools and documentation:** Developed digital layout visualizations and maintained photographic records of AS-IS and updated layout.
- **Systems setup and label logic:** Ensured consistent mapping of brand- and voltage-based labels across shelves and boxes.
- **Sourcing:** Collected materials (labels, tape, boxes) and allocated staff to workstations according to the new layout.
- **Element testing:** Conducted operator walkthroughs to confirm layout clarity, accessibility, and ergonomic compliance.

5.1.2. Integration and Test

This phase focused on integrating the proposed layout into the warehouse and validating its functionality. Items were placed according to brand and voltage, the packing station was connected to the material zone, and operator pathways were tested for efficiency. Live testing with operators assessed layout clarity, accessibility, and labeling logic, leading to refinements such as resolving bottlenecks, adjusting label positions, and rebalancing shelves. Once approved by managers and confirmed functional by operators, staff were briefed during a ramp-up phase to ensure understanding of the new organization. The completed layout map and documentation were then delivered to support training, troubleshooting, and ongoing improvement.

5.1.2.1. Implementation Plan

A structured implementation plan was developed to execute the proposed layout improvement to minimize disruption to ongoing warehouse operations. The initial plan defined tasks, responsibilities, dependencies and was scheduled to be completed in three working days.

During execution, however, several adjustments were made due to last-minute stakeholder meetings and operational constraints. Tasks were reordered, modified, and partially implemented. While the core objectives of the layout improvement were maintained, listening to stakeholder feedback during this phase improved acceptance, reduced implementation risks, and achieved a more sustainable final solution. Adopted improvements are listed in Section 5.1.2.2, while a full implementation plan comparing the initial plan and final plan is presented in Appendix C.4. Note that these changes also affected the design phase.

5.1.2.2. Prototype

Following the implementation plan, the proposed layout was realized as a physical prototype in the ReMade warehouse environment. Executed changes listed below:

- **Charger reorganization** – Chargers are moved from mixed, unlabeled bins to labeled crates sorted by brand and voltage, improving traceability and retrieval efficiency (Appendix C.5).
- **Brand-based aisle zoning** – Storage aisles are assigned to specific brands based on the inventory volume and order frequency, reducing picking errors and cognitive load while improving ergonomic access to high-demand items (Appendix C.6 – 9).
- **Standardized labeling** – Labels are repositioned and standardized to improve scanning visibility, and reduce unnecessary lifting and movement (Appendix C.10).

- **Packaging area relocation** – The packaging area is rearranged and divided into two packaging and material zones to improve workflow and reduce unnecessary movement (Appendix C.11).
- **Packing material consolidation** – Packing materials are consolidated and positioned adjacent to the packing station, creating a clear and more ergonomic packing area (Appendix C.12).
- **Distinction of inbound and outbound areas** – Inbound and outbound zones are reallocated and visually marked to restrict movements for safety and improved workflow (Appendix C.13 – 15).
- **Computer station setup** – The computer workstation is streamlined by removing unnecessary furniture and reorganizing equipment to efficient workflow (Appendix C.16).

5.1.2.3. Validation

The validation of layout improvement applied a mixed-methods approach, combining scenario-based quantitative analysis and qualitative user survey from operators. This approach enabled both objective performance measurement and subjective user perception assessment.

5.1.2.3.1. Quantitative Validation: Scenario-Based Distance Analysis

A quantitative validation method was developed to assess the impact of the proposed layout. A representative picking scenario was defined, focusing on laptop order picking & packing, as this task reflects frequent and repetitive warehouse operations. To ensure comparability between the As-Is and prototyped layout, the analysis focused on aisle-level travel paths from task initiation to completion. The analysis excluded movement within a single shelf, as it varies a lot depending on the order. However, it is notable to mention that brand-based aisle zoning clearly reduces the travel time, since some orders are single-brand orders. The movements were categorized into:

- **Value adding movement** – representing necessary travel to complete the picking task, and
- **Non-value adding movement** – representing repetitive, unnecessary back-and-forth travel.

Figure 5 represents the scenario, green and red paths indicating value-adding and non-value adding movements respectively. The quantitative analysis shows a significant improvement in operational efficiency under the prototyped layout. Total travel distance was reduced by approximately 30%, and non-value-adding movement was reduced by approximately 56%.

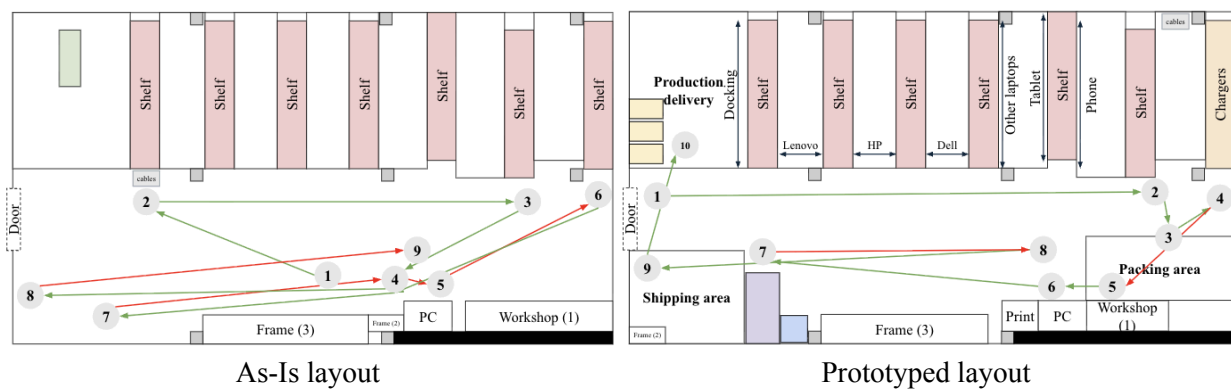


Figure 5. Visual representation of scenarios testing

5.1.2.3.2. Qualitative Validation: Staff Perception Survey

To complement the quantitative analysis, a survey-based qualitative validation was conducted to capture warehouse staff perceptions of the prototyped layout. The survey applied Likert-scale questions addressing navigation efficiency, movement reduction, workflow smoothness, safety, packing area functionality, and overall task efficiency. The full survey result is presented in Appendix C.17.

Overall, respondents reported neutral to positive perceptions of the new layout. Improvements were most notable in reduced unnecessary movement and clearer item organization, with respondents agreeing that the layout supports more efficient picking and helps maintain items in the correct locations. Perceptions of encountering misplaced items and overall workflow smoothness, including inbound and outbound processes, were more neutral, as existing items remained in place. Longer operations may further clarify the benefits of the proposed layout.

The survey findings are consistent with the results of the scenario-based distance analysis, particularly regarding reductions in unnecessary movement and improved task efficiency. Although the survey sample size ($N = 2$) is limited, this alignment between quantitative measurements and user perception provides preliminary validation of the proposed warehouse layout.

5.2. Implementation of Standard Operating Procedure

5.2.1. Element Creation

The SOP element creation is simpler than the layout improvement, since content development and drafts are developed and refined in the design phase. Thus, the element creation phase in the implement phase was applied as a checklist to evaluate whether the developed SOP followed all the necessary steps and considered everything. The general element creation phase contains hardware manufacturing, software coding, sourcing, element testing, and element refinement. Each activity was interpreted in the context of SOP creation (Table 6).

General activity	SOP development activities
Hardware manufacturing	Determination of SOP format, digital and paper form
Software coding	Document detail, procedure, safety, etc.
Sourcing	Literatures, hands-on learning, AI (image generation)
Element testing	Validation from operators, data and privacy protection
Element refinement	Feedbacks, layout changes

Table 6. SOP creation activities in general activity listed in element creation phase

5.2.2. Integration and test

After the final draft of SOP was confirmed, a user test was applied to validate the impact of the new SOP. The test involved four inexperienced participants aged between 20 and 30 years old. The participants were divided into two groups: a comparison group which followed the original SOP, and a target group which used the new SOP. Each group consisted of two participants.

The test task involved the laptop in-shelving process, identified as the primary cause of lost items. Both groups in-shelved 20 laptops under the same conditions. Cycle time, error frequency, and assistance frequency were recorded to evaluate efficiency, accuracy, and user independence. Afterward, participants completed a survey on clarity, ease of use, and overall satisfaction. Each session lasted about 30 minutes.

Results (Figure 6) show clear improvements with the new SOP. Tasks were completed 30 percent faster, with 50 percent fewer errors and 64 percent less assistance required. Survey feedback confirmed higher ratings for clarity, ease of following, visual guidance, and understanding speed. Error likelihood was higher for the original SOP, indicating participants perceived a greater chance of mistakes. Overall, both quantitative and qualitative results demonstrate that the new SOP improves efficiency, accuracy, and usability for inexperienced users.

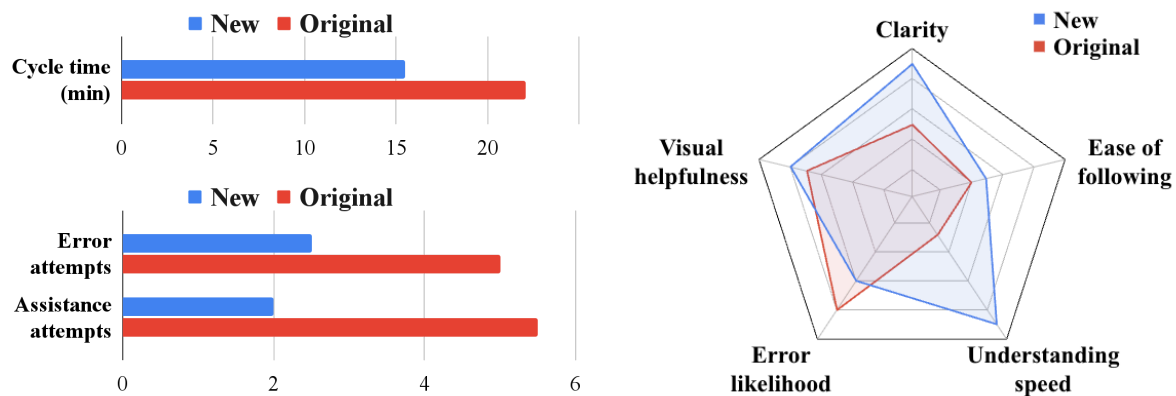


Figure 6. SOP validation result from the user test

Despite the positive results, the validation is subject to similar limitations with layout improvement, primarily related to the small sample size ($N=2$ per SOP), which limits the reliability and generalization. Therefore, the findings should be interpreted as preliminary, and further evaluation with a larger number of users is required for stronger validation.

5.3. Cost Benefit Analysis

In the implement phase, cost-benefit analysis (CBA) is conducted to evaluate the economic feasibility of the combined implementation of the layout improvement and SOP solutions. The expected benefits are calculated based on the validation results and assumptions derived from observed warehouse operations. Below lists the assumptions:

- Total 1120 hours of labor costs with student hourly payment (165 sek/hr)
- Average Swedish operator labor cost (175 sek/hr) and manager labor cost (250 sek/hr)
- Approximately 8 orders per day and minimum 30s of travel time reduction per order (minimum 27m of travel distance reduction and average operator speed of 50m/min)
- Approximately 2 items reduced in lost items per month (3000 kr per item with average 20 percent of depreciation rate) with 75% of searching time reduction (6 hrs per quarter year)

- Approximately 2 hours per month of training time reduction

Cost (SEK)		Benefit (SEK)	
Labor	184,800	Travel time	252
Seminar	2,100	Error reduction	1,700
Operator training	1,400	Training time	350
Tools	130		
Total	188,430	Total	2,302
Payback period (cost / benefit; yrs)			6.8

Table 7. Cost benefit analysis result of layout improvement and SOP solutions

Table 7 shows a financial payback period of about 6.8 years, which could be misinterpreted as a failure. However, the analysis omits key non-financial benefits, including standardization, traceability, visibility, safety, compliance, and customer satisfaction. These benefits have long-term impact by reducing process complexity and enabling future digitalization. The solution should therefore be evaluated on both financial and non-financial benefits, with the latter providing substantial long-term value.

6. Operate Phase

In this phase, practices or systems for lifecycle support and evolution of the provided solutions are suggested to facilitate long-term sustainable impact creation across the entire lifecycle of the solution. Therefore, effective lifecycle support and evolution practices for layout improvement and SOP are presented involving manufacturing philosophies, such as lean management.

6.1. Operation of Layout Improvement

6.1.1. Lifecycle Support

To provide suitable lifecycle support practices for layout improvement, its impact on other stakeholders are initially analyzed: faster and more accurate order processing in sales & distribution, streamlined workflows in operations, optimized storage and material flow in logistics, and improved fulfillment reliability and service quality for Customers.

6.1.1.1. Maintenance via 5S

To sustain the observed improvements, a weekly 5S routine was implemented. The 5S methodology (Sort, Set in Order, Shine, Standardize, Sustain) is a foundational Lean practice designed to eliminate waste, maintain workplace organization, and support continuous improvement (Hirano, 1995).

In this project, the 5S routine includes:

- Sorting and removing unnecessary items to keep workstations clear
- Setting materials in order by location and brand for easier access
- Shining and cleaning work areas, including shelves and packing zones
- Standardizing layout and labeling to ensure consistency across shifts

- Sustaining the process with reference checks and weekly verification to prevent regression

The designed 5S sheet was created in both English and Swedish to accommodate potential language barriers among operators. It includes a checklist, as well as fields for the responsible employee's name and date, allowing each operator to take turns completing the routine every Friday before the end of their shift. This sheet was introduced and handed over to operators during the final presentation at ReMade, providing a practical tool to maintain layout organization and support continuous improvement (Appendix D.1).

6.1.2. Evolution

To support adaptation of the layout, a visual continuous improvement system was proposed to identify problems and enable ongoing improvements. Grounded in Lean visual management principles from the Toyota Production System, it allows process status to be understood at a glance without verbal communication or formal meetings (Liker, 2004; Greif, 1991) and engages employees at all levels to facilitate continuous improvement (Imai, 1986).

Figure 7 presents the tool that fosters continuous improvement of the layout. Color- and symbol-coded stickers were used on the warehouse whiteboard, following the Gemba principle of solving problems at the source (Ohno, 1988). Orange, yellow, and blue indicate high, medium, and low urgency, and symbols accommodate employees with color vision deficiencies. Operators can also submit improvement ideas anonymously, reducing communication barriers and encouraging participation.

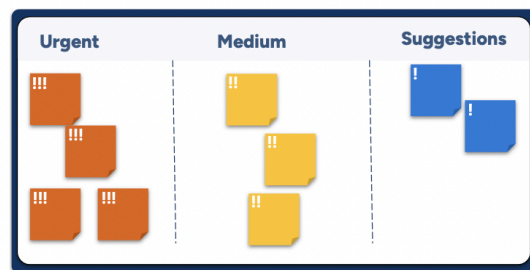


Figure 7. Example of visual system board (orange (!!!): urgent/most important, Yellow (!!): medium urgency, blue(!): future suggestions)

6.2. Operation of Standard Operating Procedure

6.2.1. Lifecycle Support

Similarly to Section 6.1.1, the impact of SOP to other stakeholders are evaluated to effectively guide the maintenance practices. For sales and distribution, reduction of lost items improve order confirmation reliability and reduce delivery delays. From an operational logistics perspective, clearer workflow instructions allow production staff to assist warehouse tasks during the peak seasons with minimum disturbance to warehouse staff. For managers, increased process visibility allows a better understanding

of current status of warehouse operations, enabling better control and decision-making across the organizations.

For the minimum complexity of maintenance practices and highest effectiveness of maintenance and improvement of SOP throughout the entire lifecycle, two sections in the SOP document, namely “*Section 6. Records and Improvement* and *Section 7. Revision History Table*”. Further explanation is continued in Section 6.2.2.

6.2.2. Evolution

Current warehouse operations in ReMade frequently change due to business expansion in scale and product diversity. Evolution is therefore critical for the SOP to adapt effectively and maintain impact. SOP evolution focuses on three aspects: **motivation**, **improvement**, and **traceability**. Motivation enables operators to take ownership by reporting issues and identifying opportunities. Improvement translates these issues into actionable plans, while traceability ensures all changes are documented and reviewable.

Two SOP sections support this evolution with minimal maintenance: *Section 6, Records and Improvement*, captures feedback, issues, and proposed actions, while *Section 7, Revision History Table*, records approved changes, version updates, and responsible authors. As the company lacks a centralized digital platform, a separate system was avoided to prevent inefficiency and complexity. Evolution and maintenance practices are integrated directly into the SOP. Figure 8 shows an example of these sections.

Motivation			How to improve		Traceability
Section 6	Date	Reported by	Description	Action	Status
	20/10/2025	Operator	Shelf numbering sequence causes long travel time	Included in layout redesign proposal	In progress
	25/10/2025	Operator	Repeated screen checks required during scanning		Open
Section 7	Version	Date	Description of change	Author	Approved
	1.0	03/11/2025	Initial Release	KTH team	

Figure 8. Example usage of Section 6 and 7 in the SOP document

7. Discussion on Governance, Sustainability and Ethical Considerations

Table 8 describes a structure of sustainability and ethical consideration. While the company mission and business model ensures the listed standards are well streamlined for their product, it was revealed from the conceive phase that there are several shortcomings in process perspective, such as process efficiency, consistency, and poor resource utilization. The layout improvement is proposed to maximize process efficiency and resource utilization. In contrast, the SOP addresses process quality, focusing on error prevention and standardization. The instructions prevent scenarios in which products can be damaged

and prevent lost items. Privacy and data protection is considered in SOP when logging in to the ReMade ERP system. Quality KPIs could not be explored due to low digital maturity of the warehouse.

Sustainability and Governance			
Sol.	9001 (Quality)	14001 (Environment)	27001 (Cyber security)
Layout	• Process efficiency • Continuous improvement	• Resource saving	• -
SOP	• Documented information • Low quality variation • Quality KPIs	• -	• Data handling • Security instruction • Authentication
Sustainability and Ethical Considerations			
1. <i>Safety, health, and welfare of society</i>	CDI	5. <i>Problem of many hands</i>	CD O
2. <i>Honesty, accuracy, and transparency</i>	CDIO	6. <i>Maintaining technical competence</i>	CDIO
3. <i>Forward looking responsibility</i>	CDIO	7. <i>Protection of sensitive information</i>	CDI
4. <i>Openness to criticism</i>	CDIO	8. <i>Long-term societal impact</i>	D O

Table 8. Governance, sustainability, and ethical considerations throughout CDIO phases

Ethical considerations are also emphasized differently across the CDIO phases. *Safety, health and welfare of society* are mainly addressed during the conceive, design, and implement phases, where physical strains and ergonomics are identified and translated into design requirements. *Honesty, accuracy, and transparency*, and *Openness to criticism* are considered throughout all phases by sharing all project materials in a common folder, conducting regular internal, course, and company presentations. Company presentations were critically valuable to receive valuable insights from operators and managers, especially when evaluating the solution feasibilities. *Forward looking responsibility* and *Problem of many hands* are considered through entire project management activities, such as WBS, risk analysis, and progress planning, ensuring clear task ownership and responsibility. In contrast, there lacked a clear line of responsibility during prototyping in the implement phase, partially contributing to delay of task completion. *Maintaining technical competence* is continuously considered by selecting the right tools for the right purposes. *Protection of sensitive information* is addressed during data collection processes, including interviews, surveys, and meetings. Lastly, *Long-term societal impact* is mainly considered during the Design and Operate phase through lifecycle support and evolution perspectives.

8. Conclusion

8.1. Discussion

The project successfully addressed key inefficiencies in ReMade warehouse operations through a combination of layout improvement and SOP. The layout improvement reduced travel distance and non-value-adding movements, leading to more efficient material flow and improved operational clarity. In parallel, the new SOP improved task execution by increasing process consistency, reducing errors, and

supporting non-expert operators. Together, these interventions resulted in operational improvements while integrating the sustainability principles.

The adopted CDIO framework significantly contributed to a successful project. Conceive phase enabled a structured understanding of ReMade warehouse operations by combining various data collection methods, which supported to establish clear problem definition and engineering requirements. Also, systematic establishment of project plans allowed the team to have forward-looking responsibility to mitigate issues without any severe operational disturbance. Design phase facilitated the systematic development and selection of the most impactful solution, allowing the selected solutions to be further refined according to the project mission and engineering requirement. Implement phase supported the realization and validation of selected solutions, ensuring feasibility and functionality. Lastly, Operate phase allowed the company to facilitate efficient lifecycle support and continuous improvement of the given solutions by taking over the practices inheriting lean philosophies.

In addition, a key success factor underpinning these outcomes was active involvement of relevant stakeholders during the entire project. Continuous input from warehouse operators and managers ensured that identified problems reflected real operational challenges, while regular feedback during design and implement phase improved solution feasibility and acceptance.

8.2. Limitations and Future Research

Several operational constraints affected the project. Limited resources and warehouse workflow alignment prevented certain planned interventions from being fully executed, such as relocating chargers to lower shelves, and expansion of SOP to other product types. Stakeholder feedback occasionally revealed misalignments in expectations, requiring flexibility in both layout modifications and SOP testing. These experiences highlighted the importance of prioritization, adaptability, and continuous stakeholder engagement, emphasizing that interventions must account for real operational constraints, including staff availability and ongoing warehouse activities, to be both feasible and effective.

The lack of systematic consideration of scalability and the use of CBA also limited the precise evaluation of proposed solutions, as non-financial benefits including visibility, safety, customer satisfaction, and scalability could not be explored. reduced As a result, while applied tools demonstrated immediate clear operational improvement, the long-term impact could not be comprehensively assessed.

To maintain and expand the benefits of the implemented solutions, it is recommended that ReMade continues structured monitoring and iterative refinement. Regular use of the provided 5S sheet is encouraged to sustain organization, cleanliness, and workflow clarity. Operator feedback mechanisms, such as the proposed visual whiteboard system, could support ongoing identification of inefficiencies and allow staff to contribute improvement suggestions. Expansion of SOPs to additional product families is advised, ensuring consistent procedures across all items handled. Finally, periodic reviews of the layout and SOP performance should be conducted to adapt to changes in resource availability, product volumes, and workflow demands, reinforcing long-term efficiency and operational resilience.

9. References

- Abhishek, P.G. and Pratap, M. (2020)** ‘Achieving Lean Warehousing Through Value Stream Mapping’, *South Asian Journal of Business and Management Cases*, 9(3), pp. 387–401.
- Agyabeng-Mensah, Y., Afum, E. and Baah, C. (2024)** ‘Stakeholder pressure and circular supply chain practices: Moderating roles of environmental information exchange capability and circular innovation orientation’, *Business Strategy and the Environment*, 33(6), pp. 5703–5720. <https://doi.org/10.1002/bse.3779>
- Barbé, L., Lombard, M., Montfort, J. and Soubeyran, Y. (2016)** ‘The art of writing and implementing standard operating procedures (SOPs) for laboratories in low-resource settings: Review of guidelines and best practices’, *Journal of Laboratory Medicine*, 40(4), pp. 229–236.
- Boardman, A.E., Greenberg, D.H., Vining, A.R. and Weimer, D.L. (2020)** ‘Efficiency without apology: Consideration of the marginal excess tax burden and distributional impacts in benefit–cost analysis’, *Journal of Benefit-Cost Analysis*, 11(3), pp. 457–478.
- CDIO Standards 2.0 (n.d.)** ‘The Context: Conceive, Design, Implement, Operate’, *Worldwide CDIO Initiative*. Available at: <https://www.cdio.org/implementing-cdio/standards/12-cdio-standards> (Accessed: 9 January 2026).
- Crawley, E.F., Malmqvist, J., Östlund, S., Brodeur, D.R. and Edström, K. (2014)** *Rethinking Engineering Education: The CDIO Approach*. 2nd edn. New York: Springer.
- De Koster, R., Le-Duc, T. and Roodbergen, K.J. (2007)** ‘Design and control of warehouse order picking: A literature review’, *European Journal of Operational Research*, 182(2), pp. 481–501.
- Dick, J., Hull, E. and Jackson, K. (2017)** ‘Requirements Engineering in the Problem Domain’, in *Requirements Engineering*. Cham: Springer International Publishing, pp. 113–134.
- Dick, J., Hull, E. and Jackson, K. (2017)** ‘Requirements Engineering in the Solution Domain’, in *Requirements Engineering*. Cham: Springer International Publishing, pp. 135–158.
- Freeman, R.E. (1984)** *Strategic Management: A Stakeholder Approach*. London: Pitman.
- Frey, D.D., Herder, P.M., Wijnia, Y. and Subrahmanian, E. (2009)** ‘The Pugh Controlled Convergence Method: Model-Based Evaluation and Implications for Design Theory’, *Research in Engineering Design*, 20(1), pp. 41–58.
- Greif, M. (1991)** *The Visual Factory: Building Participation Through Shared Information*. Portland: Productivity Press.
- Hassan, M.K., Rusli, M.H.M., Salleh, N.A.M., Kayat, S.M. and Wen-Hann, J.G. (2024)** ‘Development of Interactive Warehouse Operational Visualization’, *International Journal of Integrated Engineering*, 16(8), pp. 10–19.
- Hirano, H. (1995)** *5 Pillars of the Visual Workplace*. Boca Raton: CRC Press. Available at: <https://books.google.com/books?hl=en&lr=&id=9ObvCcJsz1kC&oi=fnd&pg=PR9> (Accessed: 9 January 2026).
- Hollmann, S., Frohme, M., Endrullat, C., Kremer, A., D’Elia, D., Regierer, B. and Nechyporenko, A. et al. (2020)** ‘Ten simple rules on how to write a standard operating procedure’, *PLOS Computational Biology*, 16(9), e1008095. <https://doi.org/10.1371/journal.pcbi.1008095>
- IEEE (2020)** *IEEE Code of Ethics*. IEEE Policies, Section 7 – Professional Activities. Available at: <https://www.ieee.org/about/corporate/governance/p7-8> (Accessed: 9 January 2026).
- Imai, M. (1986)** *Kaizen: The Key to Japan’s Competitive Success*. New York: McGraw-Hill.

- Kaplan, R.S. and Norton, D.P. (1992)** ‘The balanced scorecard: Measures that drive performance’, *Harvard Business Review*, 70(1), pp. 71–79.
- Liker, J.K. (2004)** *The Toyota Way: 14 Management Principles from the World’s Greatest Manufacturer*. New York: McGraw-Hill.
- Manghani, K. (2011)** ‘Quality assurance: Importance of systems and standard operating procedures’, *Perspectives in Clinical Research*, 2(1), pp. 34–37.
- Mitchell, R.K., Agle, B.R. and Wood, D.J. (1997)** ‘Toward a theory of stakeholder identification and salience’, *Academy of Management Review*, 22(4), pp. 853–886.
- Norheim, O.F. (2018)** ‘How can MCDA tools improve priority setting? Four critical questions’, *Cost Effectiveness and Resource Allocation*, 16(Suppl 1), p. 44.
- Ohno, T. (1988)** *Toyota Production System: Beyond Large-Scale Production*. Portland: Productivity Press.
- Rouwenhorst, B., Reuter, B., Stockrahm, V., van Houtum, G.-J., Mantel, R.J. and Zijm, W.H.M. (2000)** ‘Warehouse design and control: Framework and literature review’, *European Journal of Operational Research*, 122(3), pp. 515–533.
- Siegel, N.G. (2020)** *Engineering Project Management*. Hoboken: John Wiley & Sons.
- Siems, E., Seuring, S. and Schilling, L. (2023)** ‘Stakeholder roles in sustainable supply chain management: A literature review’, *Journal of Business Economics*, 93(4), pp. 747–775. <https://doi.org/10.1007/s11573-022-01117-5>
- Taebi, A. (2021)** *Ethics and Engineering: An Ethics-Up-Front Approach*. Unpublished manuscript.
- Tleuken, A., Roetzer, P., Fraccascia, L. and Yazan, D.M. (2025)** ‘Designing a stakeholder engagement framework with critical success factors for Hubs for Circularity’, *Journal of Environmental Management*, 384, Article 125324. <https://doi.org/10.1016/j.jenvman.2025.125324>
- Zhen, Y. and Li, X. (2022)** ‘A literature review of smart warehouse operations management’, *Operations and Supply Chain Management*, 15(4), pp. 454–469.

10. Table of Contributions

Section	Author	Section	Author
1. Introduction	Azar	4.1.2.2. Cost Analysis	Azar
2. Methodology	Azar	4.1.2.3. Pugh Concept Selection Matrix	Alara & Donghoon
3.1.1. Objectives	Alara	4.2.1. Design of Layout Improvement	Azar
3.1.2. Company Background	Azar	4.2.2. Design of Standard Operating Procedure	Donghoon
3.1.3. Business Strategy	Alara	5.1.1. Element Creation	Azar
3.1.4. Stakeholder Analysis	Alara	5.1.2. Integration and Test	Azar
3.2. Conceptual Design	Azar	5.1.2.1. Implementation Plan	Alara
3.2.1. Problem Identification and Target Problem Definition	Azar & Donghoon	5.1.2.2. Prototype	Azar
3.2.2. Engineering Requirement	Nadir & Donghoon	5.1.2.3. Validation	Alara
3.2.3. Project Planning	Donghoon	5.2. Implementation of Standard Operating Procedure	Donghoon
4.1.1. Solution Candidates	Donghoon	5.3. Cost Benefit Analysis	Donghoon
4.1.1.1. Layout Improvement	Alara	6.1.1. Lifecycle Support	Azar
4.1.1.2. Digital Dashboard	Azar	6.1.2. Evolution	Alara
4.1.1.3 Standard Operating Procedure	Donghoon	6.2. Operation of Standard Operating Procedure	Donghoon
4.1.1.4. Schedule Optimization	Nadir & Donghoon	7. Sustainability and Ethical Considerations	Nadir & Donghoon
4.1.2. Solution Evaluation	Donghoon	8. Conclusion	Azar
4.1.2.1. Satisfaction of Engineering Requirement	Donghoon		

Table 9. Table of contributions

11. Appendix A. Project Planning

Conceive			Design	Implement	Operate	Reporting
Project Formulation	AS – IS Analysis	Challenge Identification	Solution Design	Prototype Implementation	Prototype Operation	
WP	Description	Tool	WP	Description	Tool	
1. Project formulation	- Identify project objectives and success criteria - Understand company background and business	- Company background - BSC, Stakeholder analysis - WBS, RBS	4. Solution Design	- Develop potential solutions - Evaluate solutions in multiple criteria - Select prototype concept	- Architecture - Pugh concept selection matrix - MCDA	
2. AS – IS Analysis	- Observe and map existing warehouse processes - Establish stakeholder requirements - Collect data	- Business process model - Engineering requirement	5. Prototype Implementation	- Plan and build prototype - Test prototype in real-world warehouse operations - Adjust based on feedbacks	- PDCA - Literature review - Interviews/test	
3. Challenge Identification	- Evaluate key issues - Prioritize challenges - Align challenges with sustainability and goals	- Process model - Engineering requirement	6. Prototype Operation	- Monitor performance - Continuous testing and verification - Suggest long-term improvement strategies	- Kaizen - Gemba 5S checklist	

Appendix A.1. Work breakdown structure

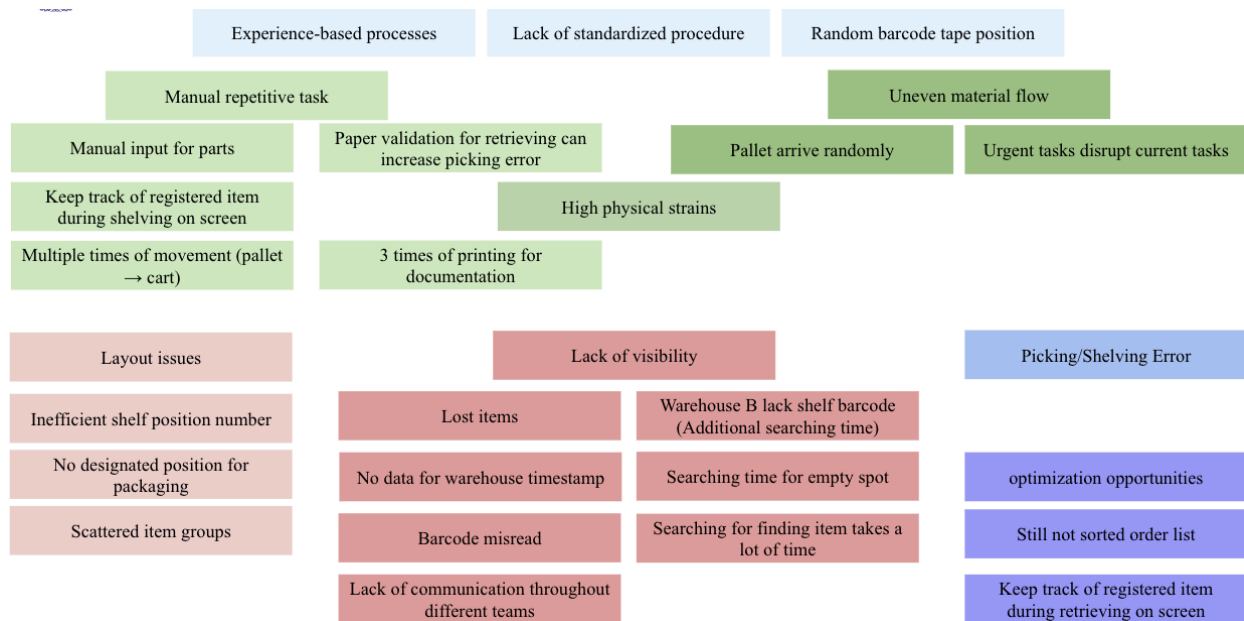
Risk Breakdown Structure (RBS)					
Level 1	Code	Level 2	Likelihood	Severity	Risk score
Management	1.1	Scope misalignment due to unclear project boundaries	1	5	5
	1.2	Lack of communication with company	1	5	5
	1.3	Lack of warehouse staff engagement	2	3	6
Conceive	2.1	Requirements not fully defined	2	4	8
	2.2	Insufficient process data collection during visits	4	3	12
	2.3	Misalignment of project goals with company needs	2	5	10
	2.4	Limited student time & resources	2	2	4
Design	3.1	Solution design lacks feasibility for small operation team	3	5	15
	3.2	Sustainability criteria not integrated	3	4	12
	3.3	Limited digital maturity for IT integration	4	2	8
	3.4	Limited student time & resources	3	3	9
Implement	4.1	Prototype disrupts warehouse operations	3	5	15
	4.2	Staff not adequately trained on new workflow	3	4	12
	4.3	Budget overruns during implementation	3	3	9
	4.4	Limited student time & resources	4	4	16
Operate	5.1	Failure to sustain improvements	3	4	12
	5.2	Poor documentation & knowledge transfer	3	3	9
	5.3	Prototype impact not measurable	3	4	12
	5.4	Limited student time & resources	4	4	16

Appendix A.2. Risk breakdown structure

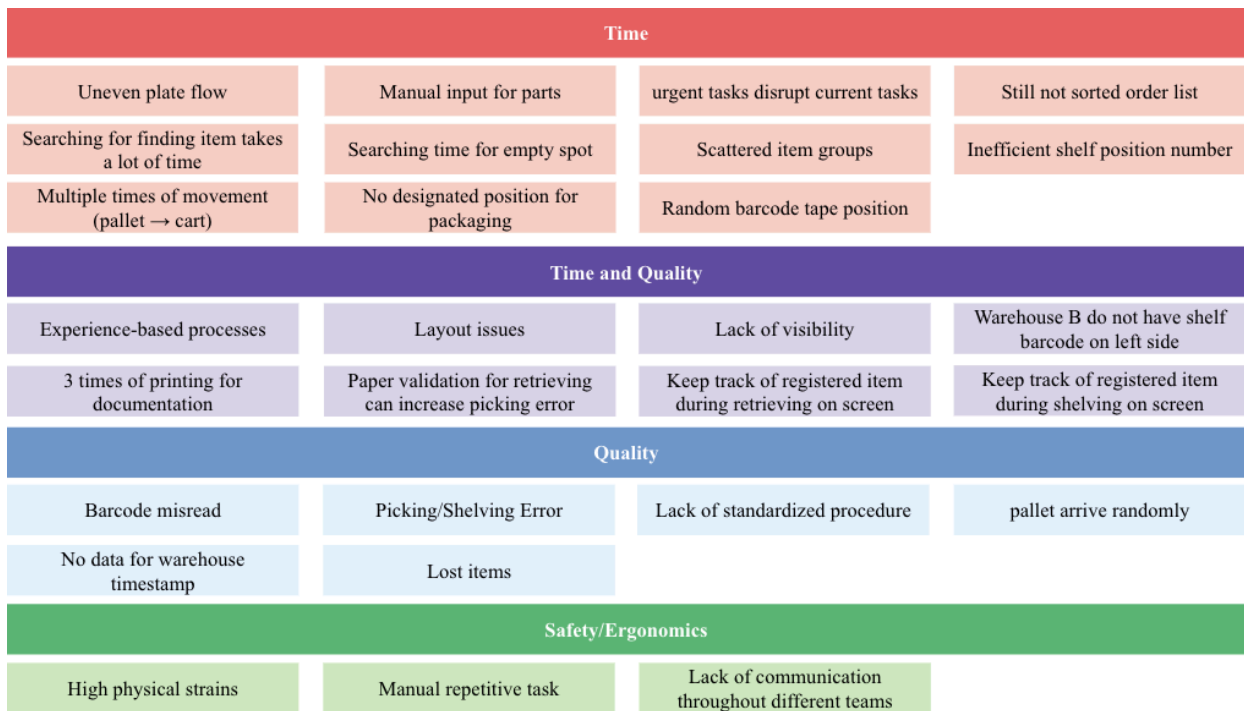
									Week																			
Phase	Work code	Task	Progress	Alara	Azar	Nad	Wes	Start date	Finish date	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	1	2	
C (400)	-	Project initiation	100%	0.0	0.0	0.0	0.0	2025.09.12	2025.09.12	09.08	09.15	09.22	09.29	10.06	10.13	10.20	10.27	11.03	11.10	11.17	11.24	12.01	12.08	12.15	12.22	12.29	01.05	
	1.1	Project mission	100%	4.0	1.5	0.0	0.0	2025.09.12	2025.09.28																			
	1.1	Risk analysis	100%	1.5	1.5	0.0	16.0	2025.09.12	2025.09.28																			
	1.1	Project plan and WBS	100%	1.5	1.5	0.0	27.0	2025.09.12	2025.09.28																			
	1.2	Company background	100%	3.0	5.5	3.0	0.0	2025.09.12	2025.09.28																			
	1.2	Business process	100%	4.0	8.0	3.0	0.0	2025.09.12	2025.10.05																			
	1.3	Business analysis	100%	11.0	4.0	3.0	0.0	2025.09.12	2025.09.28																			
	2.1-2	Value stream mapping	100%	2.0	16.0	8.0	0.0	2025.09.12	2025.10.15																			
	2.3, 3.3	Engineering requirement	100%	23.0	23.0	48.0	27.0	2025.09.12	2025.10.24																			
	2.3	Stakeholder analysis	100%	16.0	0.0	20.0	5.5	2025.09.12	2025.10.08																			
D (160)	3.1-2	Target problem definition	100%	32.0	32.0	1.5	32.0	2025.09.28	2025.10.23																			
	7.1	Conceive phase presentation	100%	5.5	5.5	8.0	16.0	2025.10.06	2025.10.16																			
	4.1	Solution architecture	100%	5.5	5.5	12.0	8.0	2025.10.16	2025.10.24																			
	4.1	Conceptual model development	100%	7.0	5.0	8.0	5.5	2025.10.16	2025.11.03																			
	4.2	Pugh concept selection matrix	100%	5.0	5.0	3.0	10.0	2025.10.20	2025.11.03																			
	4.2	Cost analysis for selected solution	100%	1.5	1.5	9.5	3.0	2025.10.20	2025.11.03																			
	4.3	Detailed design	100%	21.5	21.5	27.0	43.0	2025.10.27	2025.11.21																			
	7.1	Design phase presentation	100%	8.0	11.0	9.5	8.0	2025.10.16	2025.10.28																			
	5.1	Plan pilot implementation and Implementation	100%	31.0	33.5	27.0	35.0	2025.10.30	2025.11.24																			
	5.2	Test prototype	100%	17.0	32.0	0.0	15.5	2025.11.26	2025.12.03																			
I (240)	5.3	Prototype feedback	100%	5.0	4.0	0.0	4.0	2025.11.21	2025.12.11																			
	7.1	Implement phase presentation	100%	9.0	9.0	8.0	10.0	2025.11.21	2025.11.25																			
O (120)	6.1	Document outcomes	100%	6	24	5	10	2025.12.01	2025.12.11																			
	6.2	Suggest continuous improvement strategies	100%	8	2	6	10	2025.11.27	2025.12.11																			
	7.1	Operate phase presentation	100%	8	5	8	10	2025.12.03	2025.12.08																			
IND (120)	7.2	Final report delivery																										

Appendix A.3. Progress plan

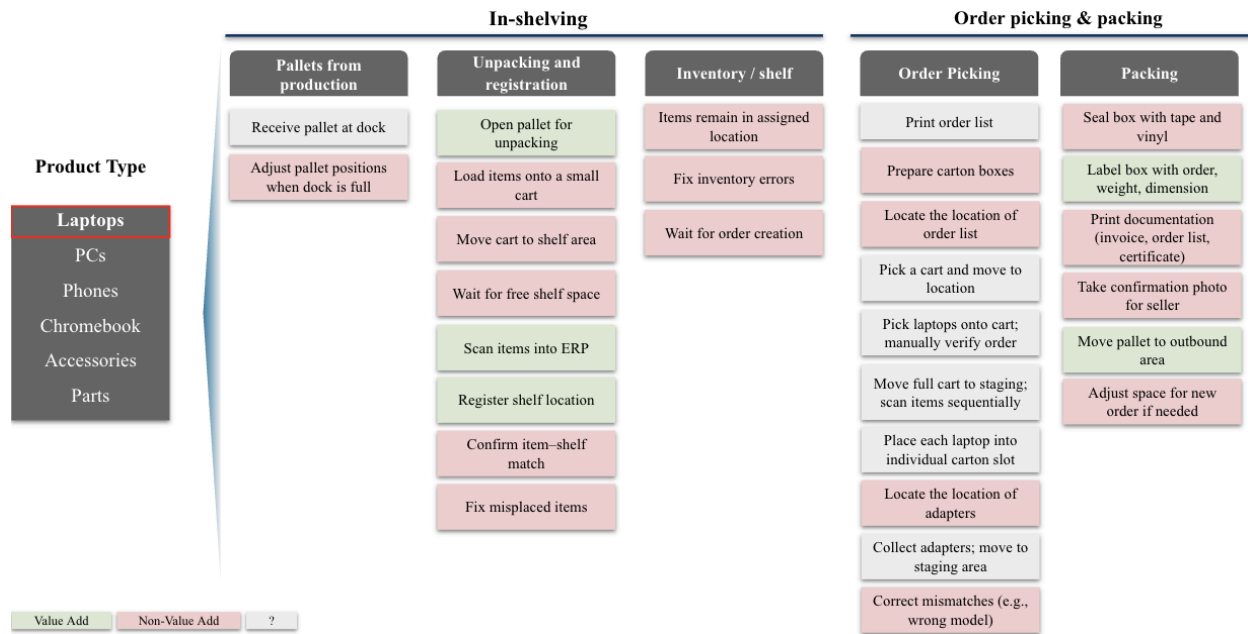
12. Appendix B. Problem Analysis and Early Design Inputs



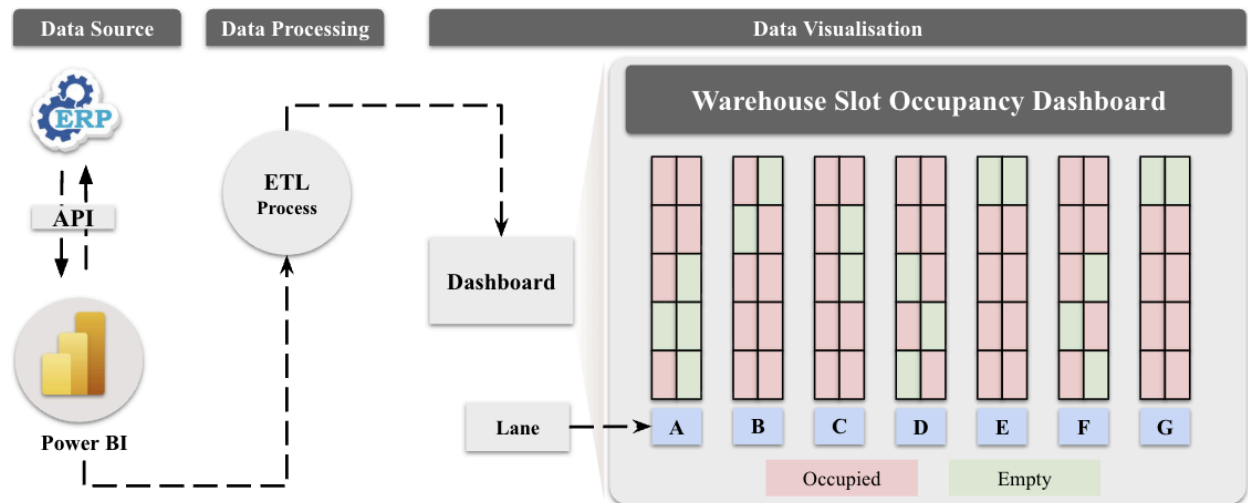
Appendix B.1. Problem group according to its theme



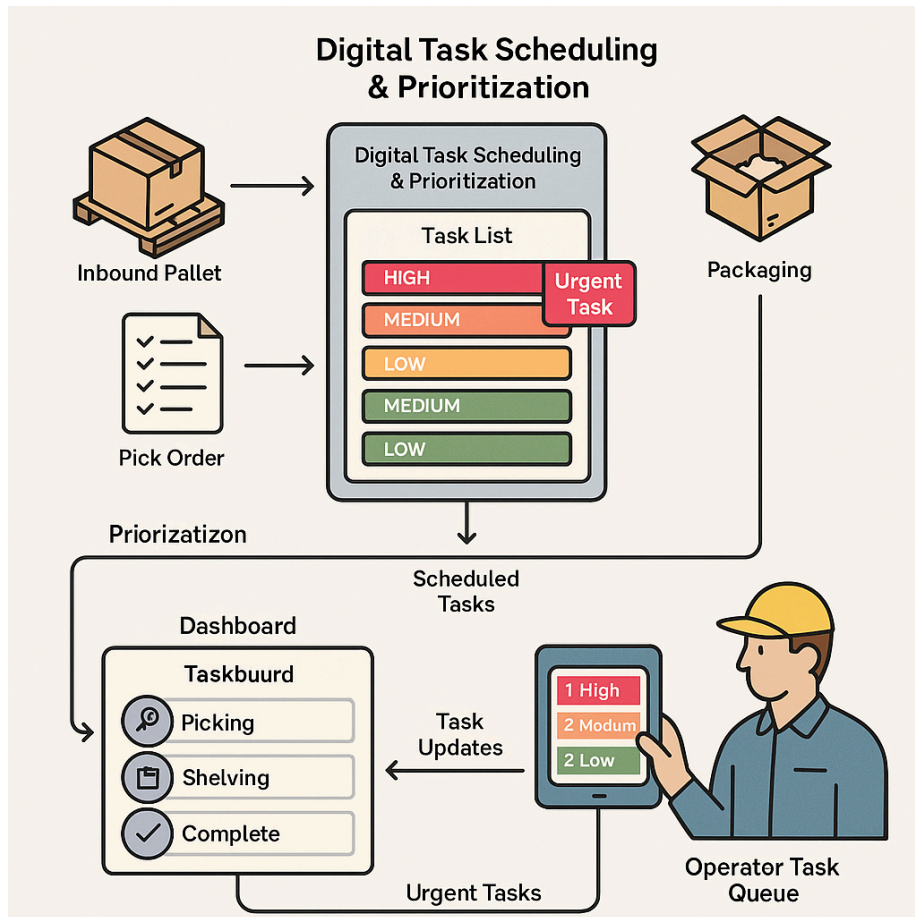
Appendix B.2. Problem group according to its impact



Appendix B.3. Summary of value stream mapping of laptop product family

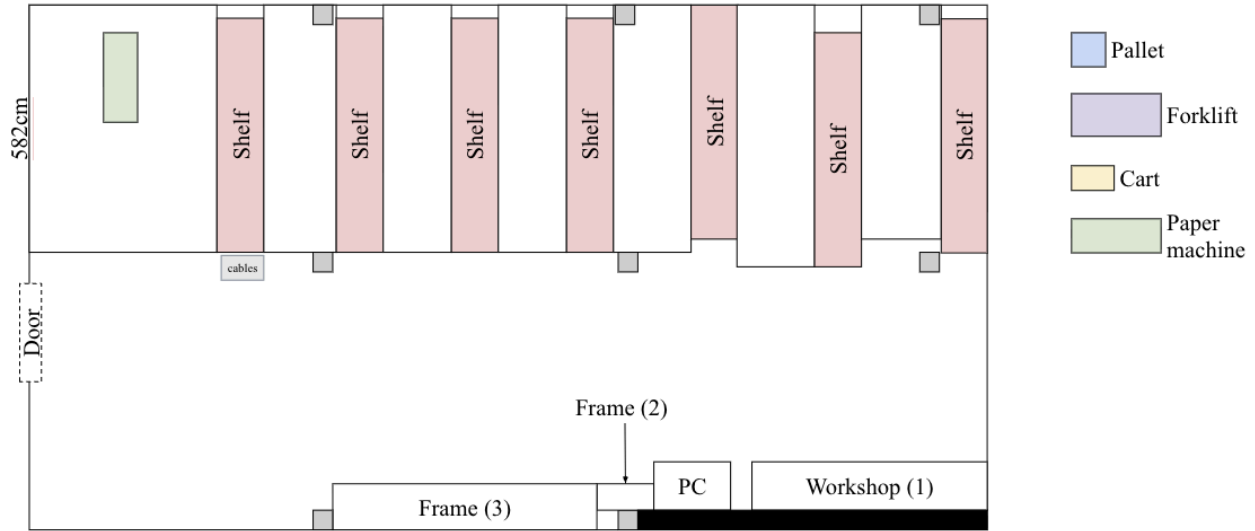


Appendix B.4. Schematic system flow of digital dashboard

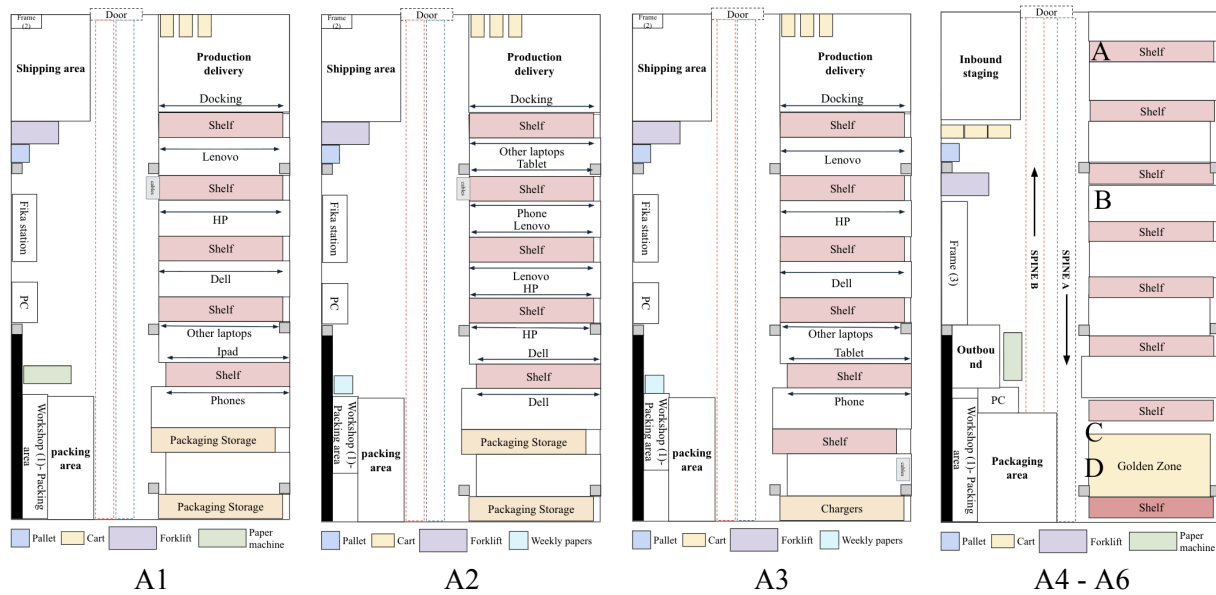


Appendix B.5. Schematic system flow of digital task scheduler (AI-generated)

13. Appendix C. Layout Improvement



Appendix C.1. As-Is layout design



Appendix C.2. Layout improvement solution, alternative A1-6

Criteria			Optimality Score						Feasibility Score						Weighted Score					
Criteria	Description	Weight	A1	A2	A3	A4	A5	A6	A1	A2	A3	A4	A5	A6	A1	A2	A3	A4	A5	A6
Time	Reducing item searching time	0.13	3	3	3	1	1	1	50	50	100	0	0	0	0.19	0.19	0.38	0.00	0.00	0.00
	Optimizing layout to minimize traveling time	0.13	5	5	3	5	1	1	50	50	100	50	0	0	0.31	0.31	0.38	0.31	0.00	0.00
Quality	Improving warehouse visibility	0.08	3	3	3	5	1	1	50	50	100	50	0	0	0.12	0.12	0.24	0.20	0.00	0.00
Safety	Designated space for packaging activities	0.08	5	5	5	5	1	3	100	100	100	50	0	50	0.40	0.40	0.40	0.20	0.00	0.12
	Minimizing items on the bottom shelf	0.08	5	5	1	1	1	3	0	0	0	0	0	50	0.00	0.00	0.00	0.00	0.00	0.12
Cost	Increase item visibility to eliminate lost products	0.08	5	5	5	1	1	1	50	50	50	0	0	0	0.20	0.20	0.20	0.00	0.00	0.00
	Minimize non-value adding time in process	0.08	3	3	3	5	5	5	50	50	50	50	50	50	0.12	0.12	0.12	0.20	0.20	0.20
															1.34	1.34	1.71	0.91	0.20	0.44

Appendix C.3. Multi-Criteria Decision Analysis

Initial Plan							Final Plan	
Day	ID	Task Name	Duration	Team	Dependencies	Notes	ID	Task Name
1	1.1	Empty shelves	3	A, B	-	Work in parallel	1.1	Feedback sessions with employees
	1.2	Barcode	3	C, D	1.1	Sort by category	1.5	Feedback session with managers
	1.3	Re-shelf initial items	3	A, B	-	Barcode must be updated	1.2	Rearrange paper machine location
	1.4	Shelf labeling	2	C, D	1.2 & 1.3	Categories + zones	1.3	Empty Workstation
	1.5	Sort unnecessary items	2	C, D	-	Alex and Neo's opinions are needed	1.4	Sort Workstation
2	2.1	Relocate Frame 3 items to packaging area and packing storage area	2	A, B	1.3	Needs to be defined before hand	2.1	Feedback session with managers
	2.2	Disassemble Frame 3	2	C, D	-	-	2.2	Re-arrange layout plan
	2.3	Move Frame 2 to new location	1	C, D	1.1	-	2.3	Arrange charger station
	2.4	Move Frame 2 + computer table	1	B	-	Check cable reach	2.4	Move packaging material from Framework 3 to workstation
	2.5	Move paper machine	1	All	-	Move carefully	2.5	Re-label charger station
	2.6	Charger station setup check- forklift, paper machine etc.	1	All	-	Cable routing	2.6	Move paper machine
	2.7	Re-shelve all sorted items	4	A, B	2.1	Match new layout	3.1	Finalise charger station
3	3.1	Start floor marking (lanes + safety)	2	D	-	Forklift + pedestrian	3.2	Finalize charger labeling
	3.2	Set up stations	2	B	-	Check ergonomics	3.3	Re-shelf ipad station
	3.3	Finish floor marking	2	D	2.5	All zones marked	4.1	Finalise charger labeling
	3.4	Packaging material storage setup	2	B	-	Bulk + daily	4.2	Finalise ipad shelving
	3.5	Full workflow test	2	All	-	Run real packing	4.3	Laptop shelf brading
	3.6	Fix layout adjustments	1	C	3.6	Small moves or tape tweaks	4.4	Laptop shelf labeling for brands
	3.7	Team training	1	All staff	3.6	Walkthrough new layout	4.5	Charger station setup check
	3.8	Video + Photo shooting for presentation	1	A	All	-	4.6	Disassemble Frame 3
							4.7	Cut & tape carton boxes
							4.8	Move computer table
							5.1	Finalise shelf labeling for brands
							5.2	Cut & tape carton boxes
							5.3	Floor marking
							5.4	Set up work station
							5.5	Photoshooting for presentation

■ Create change of plan
 ■ Fully implemented
 ■ Partially implemented
 ■ Not completed

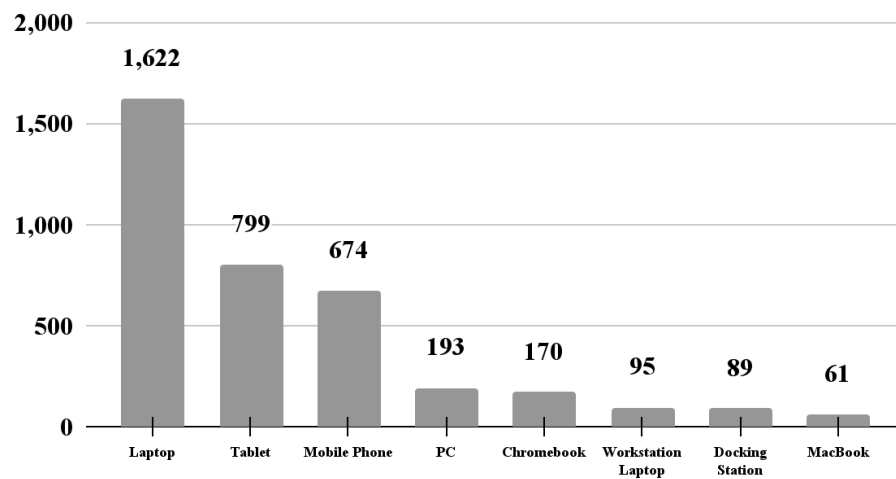
Appendix C.4. Initial and final implementation plan of layout improvement



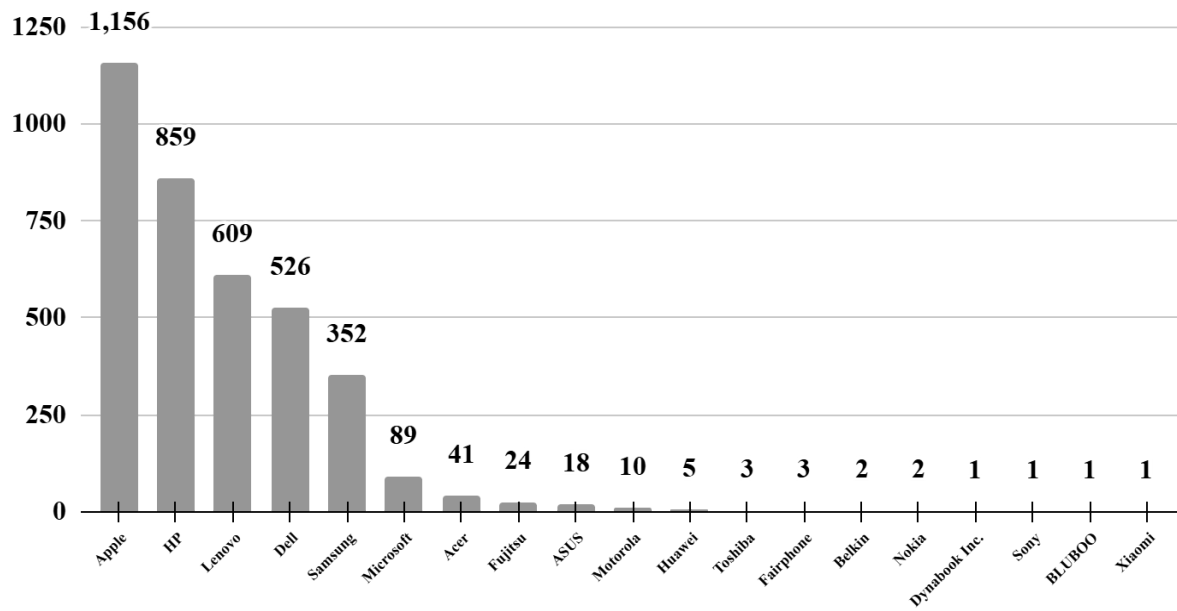
Appendix C.5. Prototype layout – charger reorganization



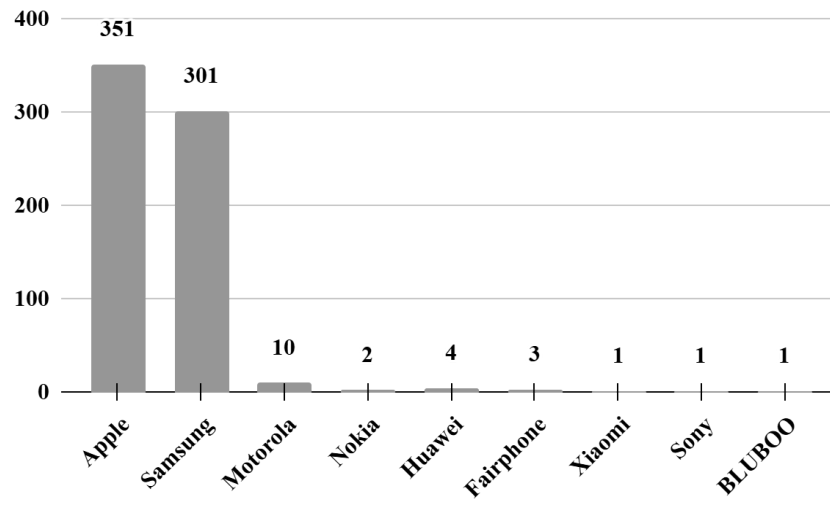
Appendix C.6. Prototype layout – brand-based aisle zoning



Appendix C.7. Total items by class in ReMade warehouse



Appendix C.8. Total Items by brand in ReMade warehouse



Appendix C.9. Total Laptops by brand in ReMade warehouse



Appendix C.10. Prototype Layout – standardized labeling



Appendix C.11. Prototype layout – packaging area relocation



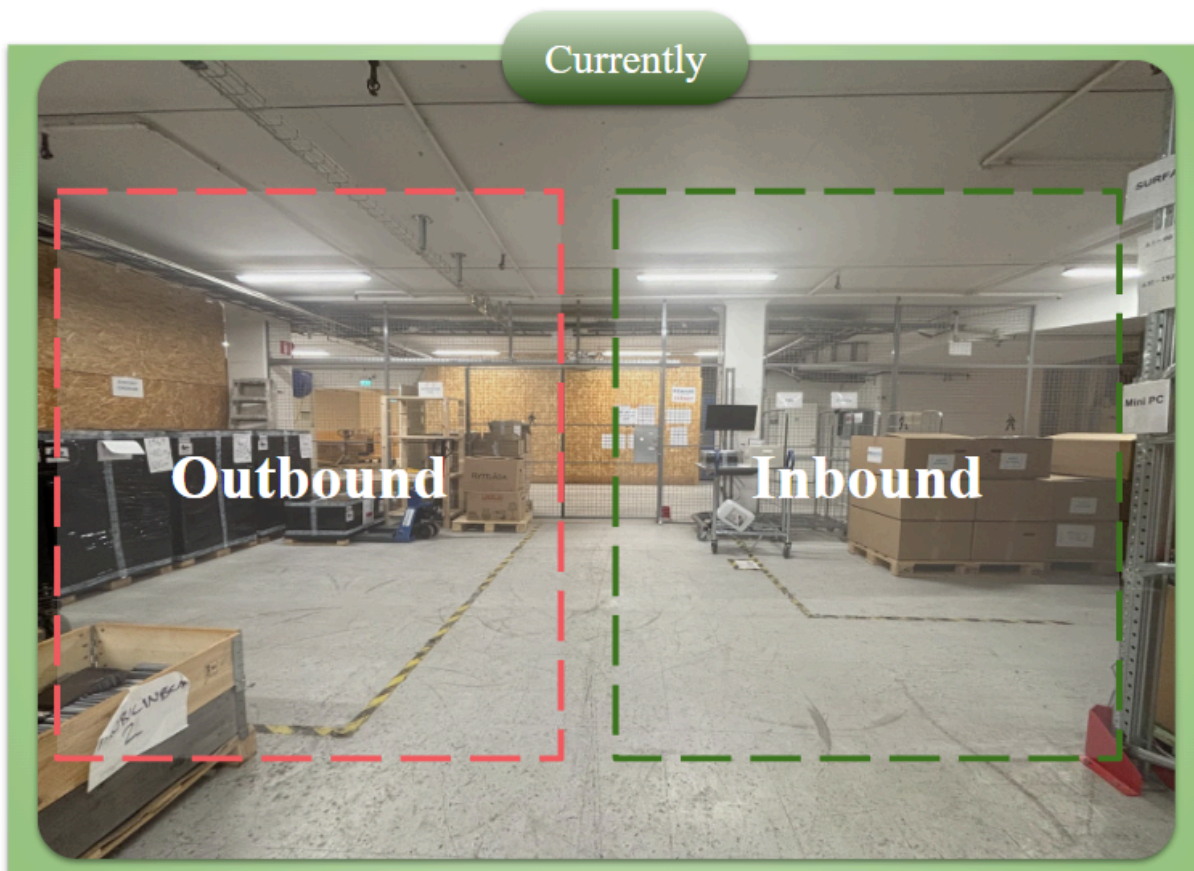
Appendix C.12. Prototype layout – packing material consolidation



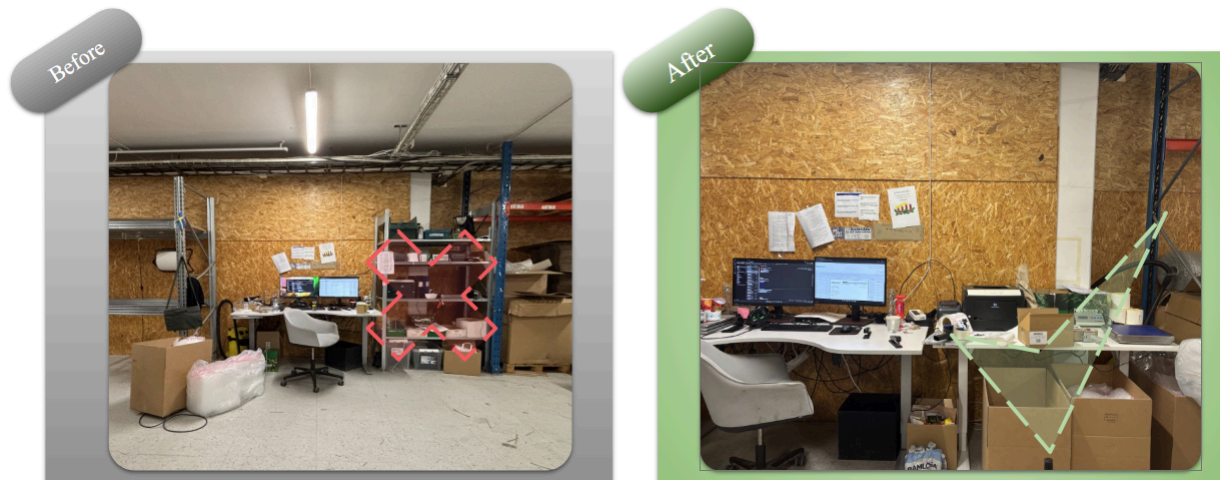
Appendix C.13. Prototype layout – inbound designated area



Appendix C.14. Prototype layout – outbound designated area



Appendix C.15. Prototype layout – overview of inbound & outbound relocation



Appendix C.16. Prototype layout – computer station setup

Questions	Employee		Mean Value
	#1	#2	
It is easier and faster to locate items in the new layout compared to the old one. (For example, brands, chargers by brand, voltage etc.)	3	4	3,5
The new layout reduces unnecessary walking or movement during picking tasks.	4	4	4
The workflow from inbound → aisle → packing station & packing material → computer station → outbound : feels smoother, with less backtracking.	3	3	3
Shelf labels and numbering are clear, readable, and easy to use.	4	3	3,5
It is easier to describe and communicate item locations to colleagues using the new layout.	5	3	4
The new layout feels safer and more organized to work in.	3	4	3,5
The new packing area reduces congestion and movement conflicts.	3	4	3,5
Items and tools at the packing station are easier to reach without strain.	4	2	3
I rarely encounter items placed in the wrong shelf or in an unexpected location.	2	3	2,5
The new layout reduces the need to double-check or redo tasks during picking or packing.	3	3	3
It is easier to keep items in the right place because the layout is more structured.	5	4	4,5
The new layout helps me avoid wasting time on unnecessary movements or confusion.	4	4	4

Appendix C.17. Survey responses of layout improvement

14. Appendix D. Tools and Supporting Materials



5S Weekly Checklist

Task	Employee name	Date
Remove empty boxes		
Clear packing table (labels, papers, stickers)		
Move misplaced items back to their correct location		
Organize items by brand/voltage again if they drifted during the week		
Reposition high-usage items to the front		
Fill the empty packaging materials		
Fill the paper box		
Wipe packing tables and laptop testing area		
Check shelves for clutter		
Ensure chargers and adapters are not tangled		
Confirm the layout map matches the real layout		
Update labels if they became unclear		
Check that all tools (tape gun, markers, cable ties) are in their assigned location		
Take a photo of the organized area as reference for next Friday		



5S Weekly Checklist

Task	Employee name	Date
Ta bort tomma lådor		
Rensa packbordet (etiketter, papper, klistermärken)		
Flytta felplacerade föremål tillbaka till rätt plats		
Organisera föremål efter märke/spänning igen om de har flyttats under veckan		
Flytta frekvent använda föremål framåt		
Fyll på de tomma förpackningsmaterialen		
Fyll papperslådan		
Torka av packbord och laptop-testområde		
Kontrollera hyllor efter ordning		
Se till att laddare och adaptrar inte är trasliga		
Bekräfta att layoutkartan stämmer med den verkliga layouten		
Uppdatera etiketter om de har blivit otydliga		
Kontrollera att alla verktyg (tejp, pennor, buntband) finns på sin tilldelade plats		
Ta ett foto av det organiserade området som referens för nästa fredag		

Appendix D.1. Operate Phase – Layout Improvement: Lifecycle Support : 5S Checklist English and Swedish Version